HUNNY TUTORIALS
YOUR CBSE PARTNER
HUNNY GUPTA
9999903256, 9999603256

HT Board Booster

Class X Science

Biology

CBSE Board Paper Question Bank

Chapter Wise Questions

Ch - 2
Control and Coordination

Session 2026-27



Class X Science Biology
Ch- Control and Coordination
Previous Year CBSE Board Question Paper

CBSE BP 2025-26 (Set- 31/1/1- 31/5/1)
Section-A (one Marker Questions)

Q.1. Human brain has various parts or regions that help in different actions, responses and coordination. From the following, identify the part responsible for precision of voluntary actions: (Set -1)
(a) Cerebrum (b) Cerebellum (c) Medulla (d) Pons

Q.2. **Assertion (A):** Reflex actions do not involve thinking. (Set -1)
Reason (R): Most reflex actions are controlled by the spinal cord.

Q.3. Identify the type of tropic movement that is responsible for the movement of the pollen tube towards an ovule from the following options. (Set -2)
(a) Hydrotropism (b) Geotropism
(c) Chemotropism (d) Phototropism
presence of starch.

Q.4. Walking in a straight line and riding a bicycle are activities which are possible due to coordination of which part of the brain? (Set -3)

(Part of the Brain)	**(Name of the area of the Brain)**
(A) Fore-Brain	Cerebrum
(B)Mid-Brain	Cerebellum
(C)Hind-Brain	Cerebellum
(D) Hind-Brain	Medulla

Q.5. In a synapse, the chemical signal is transmitted from : (Set -3)
(a) Dendritic end of one neuron to axonal end of another neuron.
(b) Axon to cell body of the same neuron.
(c) Cell body to axonal end of the same neuron.
(d) Axonal end of one neuron to dendritic end of another neuron.

Q.6. The growth of tendrils in grapes plants is due to : (Set -3)
(a) Effect of light (b) Effect of gravity
(c) Rapid cell division in tendril cells in contact with the support
(d) Rapid cell division in tendril cells that are away from the support

Q.7. Following are the statements about the different parts of the brain. Select the correct statements. (Set -4)
(i) The thinking part of the brain is the forebrain.
(ii) Centres for smell, hearing, memory, sight are located in forebrain.
(iii) Salivation, vomiting, blood pressure are controlled by medulla in the hind brain.
(iv) Cerebellum does not control the posture and balance of the body.
(a) (i), (ii) and (iii) (b) (ii), (iii) and (iv)
(c) (i), (ii) and (iv) (d) (i), (iii) and (iv)

Q.8. Identify the endocrine gland that regulates the growth of human body. (Set -5)
(a) Pituitary gland (b) Thyroid gland
(c) Pancreas (d) Adrenal gland

Section-B (Two Marker Questions)

Q.9. Give differences between the following: (Set -1)
(a) Nephron and neuron (b) Sensory nerve and motor nerve

Hunny Gupta 9999903256

Q.10. "The timing and the amount of hormone secreted by a gland are regulated in the human body." Justify this statement with the help of an example. (Set -2)

OR

Electrical impulse is an excellent means to transmit information in the animal body, but there are some limitations to the use of

Q.11. Varun is diabetic. His latest ultrasound reports showed a blockage in his pancreatic duct. How will this affect the dual functioning of the pancreas? (Set -4)

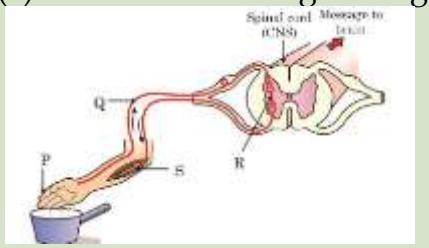
OR

Sunil unconsciously touched the hot iron rod and immediately withdrew his hand. Which type of action is involved? Also define this action. Draw a flow chart to show the path followed for this action.

Q.12. Give reasons for the following: (Set -4)

(a) Adrenal gland is known as the stress gland of the body.

Q.13. (a) Observe the given figure and identify the labelled parts P, Q, R and S: (Set -5)



OR

Which of the plant hormones are responsible for the following processes?

(i) Promote cell division (ii) Inhibition of growth

(iii) Detection of light (iv) Wilting of leaves

(b) Which of the plant hormones are responsible for the following processes ?

(i) Promote cell division (ii) Inhibition of growth

(iii) Detection of light (iv) Wilting of leaves

Section-C (Three Marker Questions)

Q.14. Read the following passage and answer the questions that follow: (Set -2)

A middle-aged person is facing some cognitive changes in himself. His early symptoms included mild forgetfulness. For example, forgetting where he left his mobile or failing to find words to complete his sentences. He was facing problems in driving a car, climbing up and down the stairs, inserting a plug into an outlet. He slowly developed a sad mood and anxiety.

(a) What are voluntary actions?

(b) Which part of the brain is responsible for precision of voluntary actions?

(c) (i) Explain the role of the medulla.

OR

(c) (ii) How do the animal muscles move?

Hunny Gupta 9999903256

CBSE BP 2024-25 (Set- 31/1/1- 31/6/1)

Section-A (one Marker Questions)

Q.15. The plant hormone whose concentration stimulates the cells to grow longer on the side of the shoot which is away from light is: (Set- 31/1/1)

(a) Cytokinins (b) Gibberellins (c) Adrenaline (d) Auxins

Q.16. Observe the given figures A and B. When chhui-mui (sensitive) plant is touched, its leaves fold. This is due to: (Set- 31/2/1)



(a) Hormonal effect (b) Thermal effect
(c) Change in amount of water in cell
(d) Electromagnetic effect

Q.17. The part of the brain which maintains the posture and balance of the body is: (Set- 31/3/1)

(a) Pons (b) Cerebrum (c) Cerebellum (d) Medulla

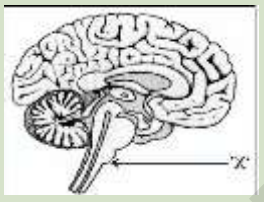
Q.18. The plant hormone present in greater concentration in the areas of rapidly dividing cells is: (Set- 31/3/1)

(a) Auxin (b) Cytokinins (c) Gibberellins (d) Abscisic

Q.19. **Assertion (A):** In our actions of writing or talking, our nervous system communicates with the muscles. (Set- 31/3/1)

Reason (R): Cranial nerves and spinal nerves form the peripheral nervous system.

Q.20. +Which among the following is not a neural action controlled by the part of human brain labelled 'X' in the figure above? (Set- 31/5/1)



(a) Salivation (b) Hunger (c) Vomiting (d) Blood Pressure

Q.21. The plant hormones promoting rapid cell division in seeds and wilting of leaves respectively are (Set- 31/5/1)

(a) Auxins and Abscisic acid (b) Cytokinins and Abscisic acid
(c) Gibberellins and Auxins (d) Abscisic acid and Gibberellins

Q.22. A pair of endocrine glands located in the human brain is: (Set- 31/6/1)

(a) Parathyroid and Pituitary (b) Pineal and Thymus
(c) Hypothalamus and Thymus (d) Hypothalamus and Pineal

Section-B

Q.23. Name the parts of hind-brain of the human brain. Which part of the hind-brain controls involuntary actions such as blood pressure and salivation? (Set- 31/2/1)

Q.24. (a) How is brain protected in our body? (Set- 31/6/1)

(b) A doctor finds in one of his patients that he is not maintaining a proper posture and balance of his body. State the region of brain and also the part of brain which is responsible for it.

Section-C

Q.25. Plants have neither a nervous system nor muscles, even then they respond to stimuli. For example, the leaves of chhui-mui (touch-me-not) plant when touched begin to fold up and droop. (Set- 31/1/1)

(a) How is the information communicated in "touch-me-not" plants?

(b) What enables the plant cells to bring out the observable response?

Hunny Gupta 9999903256

(c) Differentiate the movement mentioned above from the movement of tendrils in a pea plant.

Q.26. (a) Define hormone. (Set- 31/2/1)

(b) "Hormones should be secreted in precise quantities. We have a feedback mechanism through which this is done." With the help of an example justify the statement.

Q.27. State two limitations of electrical impulses in multicellular organisms. Why is chemical communication better than electrical impulses as a means of communication between cells in multicellular organisms? (Set- 31/4/1)

Section-D

Q.28. (A) (a) Analyse the given situations and interpret the possible reason for each:

(i) Iodine deficiency in diet increases the possibility of a disease of swollen neck in a person. (Set- 31/5/1)

(ii) Some people in population may have very short heights (dwarfs).

(iii) Thick facial hairs develop in boys at the age of 10-12 years.

(b) Explain two reasons which necessitate the need of chemical communication in multicellular organisms.

OR

(B) (a) Differentiate between voluntary and involuntary action.

(b) Define reflex action. With the help of a flow diagram, show the correct sequence of path of Nerve impulse from place of its origin.

Section-E

Q.29. A person while climbing up a rocky hill comes into a panic state and fear. His body starts reacting in a "flight-or-flight" condition to adjust to the dangerous and stressful situation. **Based on the above facts, answer the questions that follow.** (Set- 31/3/1)

(a) (i) Name the hormone secreted in the blood of the person in this situation.

OR

(a) (ii) Name the source gland of the hormone secreted in this condition.

(b) State any two responses in the body of the person as a result of the secretion of this hormone.

(c) How does the action of the chemical signal in terms of hormones differ from the electrical impulses via nerve cells?

Q.30. In life there are certain changes in the environment called 'stimuli' to which we respond appropriately. Touching a flame suddenly is a dangerous situation for us. One way is to think consciously about the possibility of burning and then moving the hand. But our body has been designed in such a way that we save ourself from such situations immediately.

(i) Name the action by which we protect ourself in the situation mentioned above and define it. (Set- 31/4/1)

(ii) Write the role of (a) motor and

(b) relay neuron.

(iii) (A) What are the two types of nervous system in human body? Name the components of each of them.

OR

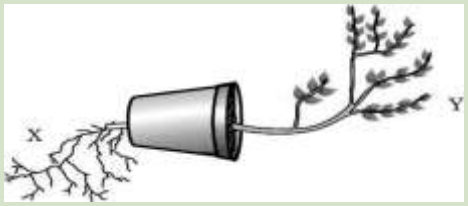
(iii) (B) Which part of the human brain is responsible for:

(a) thinking (b) picking up a pencil

(c) controlling blood pressure (d) controlling hunger

Hunny Gupta 9999903256

Q.31. The growth movements of plant parts in which the direction of the stimulus determines the direction of the response is known as tropic movements or tropism. Plants also have non-directional movements which may not be growth dependent. (Set- 31/6/1)



(I) Name the movement which causes 'X' and 'Y' to grow downwards and upwards respectively. (Refer above figure)

(II) Write the name of a hormone that plays a major role in (i) falling of leaves (ii) rapid cell division

(III) (a) Leaves of the sensitive plant move very quickly in response to 'touch'. How is this stimulus of touch communicated and explain how the movement takes place.

OR

(III) (b) Name the plant hormone which is synthesized at the shoot tip. How does this hormone helps the plant to bend towards light?

Hunny Gupta 9999903256

CBSE BP 2023-24 (Set- 31/1/1- 31/5/1)

Section-A (one Marker Questions)

Q.32. In a nerve cell, the site where the electrical impulse is converted into a chemical signal is known as: (Set- 31/1/1)

(a) Axon (b) Dendrites
(c) Neuromuscular junction (d) Cell body

Q.33. Select out of the following a gland which does NOT occur as a pair in the human body:

(a) Pituitary (b) Ovary (c) Testis (d) Adrenal

(Set- 31/2/1)

Q.34. Select from the following a plant hormone which promotes cell division. (Set- 31/3/1)

(a) Gibberellins (b) Auxins
(c) Abscissic Acid (d) Cytokinins

Q.35. A plant growth inhibitor hormone which causes wilting of leaves is called: (Set- 31/4/1)

(a) Auxin (b) Cytokinin
(c) Abscisic acid (d) Gibberellin

Q.36. The correct sequence of events when someone's hand touches a hot object unconsciously:

(a) Receptors in skin → Motor neuron → Relay neuron → Sensory neuron → Effector muscle in arm

(b) Receptors in skin → Relay neuron → Sensory neuron → Motor neuron →Effector muscle in arm

(c) Receptors in skin → Sensory neuron → Relay neuron → Motor neuron →Effector muscle in arm

(d) Receptors in skin → Sensory neuron → Effector muscle in arm → Motor neuron →Relay neuron (Set- 31/5/1)

Q.37. Sense organ in which olfactory receptors are present is : (Set- 31/5/1)

(a) Nose (b) Skin (c) Tongue (d) Inner ear

SECTION B

Q.38. (a) How is the movement of leaves of a sensitive plant different from the downward movement of the roots? (Set- 31/1/1)

OR

(b) There is a hormone which regulates carbohydrate, protein and fat metabolism in our body. Name the hormone and the gland which secretes it. Why is it important for us to have iodised salt in our diet?

Q.39. Where are auxins synthesized? How do they promote phototropism? (Set- 31/3/1)

SECTION C

Q.40. Define reflex action. With the help of a flow chart show the path of a reflex action such as sneezing. (Set- 31/1/1)

Q.41. (a) List two constituents of Central Nervous System (CNS). How are these components protected from injuries? (Set- 31/3/1)

(b) Write two limitations of the use of electrical impulses.

Q.42. Taking the example of any two animal hormones along with their gland of secretion, explain how these hormones help (i) in growth and development and (ii) regulate metabolism, in the body. (Set- 31/5/1)

Hunny Gupta 9999903256

SECTION D

Q.43. (i) Distinguish between hormonal co-ordination in plants and animals. (Set- 31/2/1)

(ii) Which part of the brain is responsible for -

(1) intelligence (2) riding a bicycle (3) vomiting

(4) controlling hunger

(iii) How is brain and spinal-cord protected against mechanical injuries?

OR

What are tropic movements? Give an example of a plant hormone which (1) inhibits growth and (2) promotes cell division.

(ii) Explain directional movement of a tendril in pea plant in response to touch. Name the hormone responsible for this movement.

Q.44. (a) (i) Define a reflex arc. Why have reflex arcs evolved in animals ? Trace the sequence of events which occur, when you suddenly touch a hot object.

(ii) Name the part of nervous system which helps in communication between the central nervous system and other parts of the body. What are the two components of this system ? (Set- 31/4/1)

OR

(b) (i) Leaves of chhui-mui plant begin to fold up and droop in response to a stimulus. Name the stimulus and write the cause for such a rapid movement. Is there any growth involved in the movement ?

(ii) Define geotropism in plants. What is meant by positive and negative geotropism ? Give one example of each type.

Hunny Gupta 9999903256

CBSE BP 2022-23 (Set- 31/1/1- 31/6/1)

Section-A (one Marker Questions)

Q.44. Select from the following the correct statement about tropic movement in plants:

(a) It is due to stimulus of touch and temperature.

(b) It does not depend upon the direction of stimulus received.

(c) It is observed only in roots and not in stems.

(d) It is a growth related movement. (Set- 31/1/1)

Q.45. In plants the role of cytokinin is: (Set- 31/4/1)

(a) Promote cell division (b) Wilting of leaves.

(c) Promote the opening of stomatal pore. (d) Help in the growth of stem.

Q.46. The part in which gustatory receptors are present in our body is: (Set- 31/5/1)

(a) Inner ear (b) skin (c) tongue (d) inner lining of nose

Q.47. Walking in a straight line and riding a bicycle are the activities which are possible due to a part of the brain. Choose the correct location and name of this part from the given table:

	Part of the Brain	**Name**
(a)	Fore brain	Cerebrum
(b)	Mid brain	Hypothalamus
(c)	Hind brain	Cerebellum
(d)	Hind brain	Medulla

(Set- 31/6/1)

SECTION B

Q.48. (a) Write the role of insulin in regulating blood sugar levels in human body. Mention the disease caused due to it. (Set- 31/1/1)

(b) How is the timing and the amount of release of insulin in the blood regulated?

Q.49. (A) Name the part of brain which is responsible for the following actions:

(i) Maintaining posture and balance (ii) Beating of heart (Set- 31/4/1)

(iii) Thinking (iv) Blood pressure

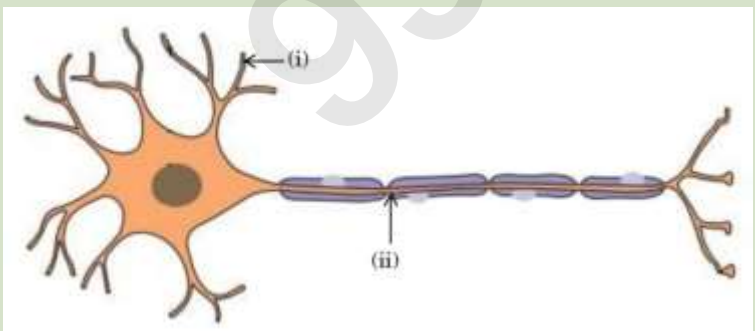
OR

(B) Where are auxins synthesized in a plant? Which organ of the plant shows:

(i) Positive phototropism (ii) Negative geotropism

(iii) Positive hydrotropism

Q.50. Write the name and function of parts (i) and (ii) in the diagram of a neuron given below.



(Set- 31/5/1)

Q.51. (a) List two differences between the movement of leaves of a sensitive plant and the movement of a shoot towards light. (Set- 31/6/1)

OR

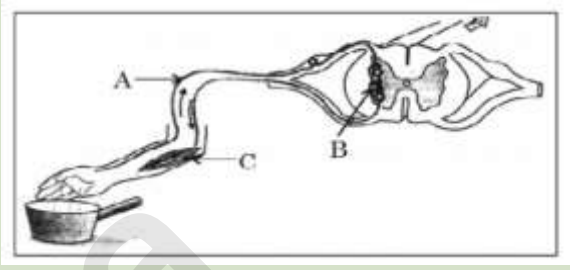
(b) What happens at synapse between two neurons? State briefly.

Hunny Gupta 9999903256

SECTION C

Q.52. Name the hormone released and the gland which secretes it in human beings during scary situations. How does the body respond to enable it to deal with the situation? (Set- 31/2/1)

Q.53. (a) Name the gland and the hormone secreted by it in scary situations in human beings. List any two responses shown by our body when this hormone is secreted into the blood.



OR

(b) In the given diagram

(i) Name the parts labelled A, B, and C.

(ii) Write the functions of A and C.

(iii) Reflex arcs have evolved in animals? Why? (Set- 31/6/1)

SECTION D

Q.54. (a) (i) List three points of difference between nervous and hormonal mechanisms for control and coordination in animals. (Set- 31/2/1)

(ii) How are auxins related with the bending of plant shoot towards unidirectional light? Explain.

OR

(b) (i) Name the disorder which a person is likely to suffer from due to the following:

(I) Over-secretion of growth hormone

(II) Deficiency of oestrogen in females

(III) Less secretion of thyroxine Also name the gland that secretes each of the hormones mentioned above.

(ii) How is the timing and amount of hormone released regulated? Explain with the help of an example.

Hunny Gupta 9999903256